Seung-yong Lee (이승용)

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EDUCATION

Sejong University, B.S. in Computer Science and Engineering – Seoul, South Korea Mar 2020 – Aug 2026

- Cumulative GPA: 4.36/4.5 — Major GPA: 4.44/4.5
- Cumulative Department Rank: 3/211 (top 1.4%)
- Early graduation after 7 semesters with *Summa Cum Laude*.

EXPERIENCE

NAVER Corp., Software Engineer Intern – NAVER 1784, Seongnam July 2025 – Aug 2025

Distributed systems research and development for real-time collaboration.

- **Skills:** Go, gRPC, MongoDB, Kubernetes, k6, CRDT (Conflict-free Replicated Data Types)
- Resolved data consistency issues caused by Yorkie API contention under concurrent workloads. (yorkie#1414)
- Defined benchmark workloads to measure and analyze server performance bottlenecks under multi-user real-time collaborative editing scenarios. (yorkie#1366)
- Defined the Document-Client state transition model to improve system reliability and predictability. (yorkie#1431)
- Resolved real-time synchronization errors caused by Pub/Sub event loss under unstable network conditions. (yorkie-js-sdk#1054, pair)
- Resolved a Text CRDT deletion defect caused by VersionVector errors. (yorkie#1437)
- Designed a DB-to-in-memory migration architecture to reduce presence interaction overhead. (yorkie#1469)
- Implemented and optimized multi-document presence retrieval in a sharded cluster environment. (codepair#498, yorkie#1391, yorkie#1402)

OPEN SOURCE CONTRIBUTIONS

Yorkie, Maintainer July 2025 – present

An open-source document store for building real-time collaborative applications.

- **Skills:** Go, gRPC, MongoDB, EKS, ArgoCD, CRDT (Conflict-free Replicated Data Types)
- Contribute to server and EKS cluster maintenance, issue triaging, and pull request reviews.
- Maintain the broader Yorkie ecosystem, including CodePair and Wafflebase, real-time collaborative tools.
- Implemented bulk retrieval for the GetDocuments API, improving performance by 14.03x. (yorkie#931)
- Identified an idempotency defect in a non-atomic distributed synchronization workflow under partial failures and defined a reproducible test scenario. (yorkie#1001, yorkie#1000)
- Optimized real-time editing performance by eliminating unnecessary database queries in key user scenarios. (yorkie#974)
- Improved bottleneck observability by exposing goroutine count metrics for individual background tasks. (yorkie#963)
- Identified and mitigated an SSRF vulnerability in the webhook feature through input validation. (yorkie#1650)

Grafana, Contributor

The open and composable observability and data visualization platform.

- Updated outdated Prometheus integration documentation to reflect recent changes. (grafana#91852)

PROJECTS

VLM-Based Dashcam Video Reasoning for Traffic Accident Liability Assessment

Capstone research project on traffic accident liability reasoning from dashcam footage.

- Designed a three-stage Vision-VLM-RAG pipeline to extract accident context, reason about traffic scenes, and retrieve relevant liability guidelines.
- Developed a motion compensation pipeline using background optical flow and RANSAC to estimate camera motion, then corrected vehicle bounding box displacement against the background to infer motion direction.
- Proposed and implemented video overlays visualizing vehicle bounding boxes and inferred motion directions to provide explicit spatial cues for VLM reasoning.
- Compared with raw video using the same VLM, overlay-enhanced video improved accident feature inference by 14.4% and vehicle direction inference by 11.2%.

ACTIVITIES & LEADERSHIP

Open Source Contribution Academy (OSSCA) July 2024 – present

Lead Mentor (2026), Mentor (2025), Lead Mentee (2024)

- Mentored open-source contributors in the Yorkie project through OSSCA, a nationwide open-source mentoring program hosted by Korea's Ministry of Science and ICT.
- Delivered a seminar on operation-based CRDTs.

GREEDY Aug 2024 – 2026

Founder, software engineering club at Sejong University

- Founded and grew a software engineering club to 40+ active members and a 200+ member open community.
- Led GREEDY to become an officially recognized student organization in 2026.
- Mentored undergraduate students through software engineering studies and code reviews.
- Organized a university-funded developer conference with 260 participants.

Woowa Tech Course Feb 2023 – Nov 2023

Backend Track, 5th Cohort

- Completed an intensive software engineering program run by Woowa Brothers, a leading Korean food delivery company.
- Delivered a technical presentation on JDBC.

HONORS AND AWARDS

- **National Excellence Scholarship for Science & Engineering**, Ministry of Science and ICT (2025 – 2026)
- **Academic Excellence Scholarship**, Sejong University (2024 – 2025)
- **President's Award for Open Source Contributions**, Korea IT Business Promotion Association (2024)